Three-Dimensional Head Pose Estimation Using a Stereo Camera Arrangement

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ABSTRACT
Head-pose estimation is a crucial component for analysing human behaviour through various 2D and 3D applications. However, the usage of the strategies based on 2D technologies is not very effective, as the sources of data are limited. In contrast, the usage of the strategies based on 3D technologies is a promising area. The 3D HPE methods are also imperfect, especially when they are applied in situations of inconsistent illumination or occlusion. An analysis of related works helps to establish an appropriate framework for the current research, and this paper extends previous work by creating an algorithm that further improves the framework. The method relies upon a stereo camera arrangement and utilises a method to detect and track key landmark points of the human face to evaluate improvement in 3D-head-pose estimation.

CCS Concepts
• **Information systems→Information systems applications**

Keywords
3D Face tracking; head-pose estimation; face detection; real-time tracking; facial landmark tracking; stereo camera; camera calibration.

1. INTRODUCTION
Head pose estimation (HPE) has always been viewed as a key element for analyzing human behavior; thus, the development of 2D and 3D applications for automatic and precise head pose analysis has profoundly benefitted such analyses [1]-[4].

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HPE has acquired exceptional significance in such areas as video surveillance, human-computer interaction, and human behavior analysis. Moreover, efficient HPE is considered an important contributor to constructive gaze tracking, face recognition, and facial expression analysis [3]-[6]. Despite considerable improvements in the field, HPE and localization of facial landmarks still pose challenges, as the majority of data is acquired from low-resolution surveillance videos and thus appears blurred. Higher image quality is now achieved via fusion of information from multiple cameras, but HPE methods regarding multi-view settings and computing the head pose of a rotating person are still scarce [7]. HPE from a single 2D image has proven challenging because of scarcity of data, which has been addressed by using either multiple 2D images or video sequences. Usually, 2D HPE is conducted with the help of appearance-based and featurebased approaches [8]. The insufficiency of 2D imaging has partly been resolved with 3D HDE methods, but the problem with 3D HPE is in the impropriety of large 3D scanners used for research and industrial purposes for broader consumer aims. A viable alternative is the use of depth cameras, but they provide low quality of analysis and require more precision [9]-[11]. Traditionally performed on RGB images with the use of rotation-specific classifiers or facial features, 3D HPE is still imprecise in situations with variation of illumination, shadows, and occlusions [3], [12]. Hence, further research is needed for effective 3D HDE method development and testing of existing techniques for the sake of advancing the quality and precision of head pose analysis. In this paper, we present a performance analysis of the paper by Alqahtani et al. [13]. to show the improvement in 3D HPE. The rest of paper is organized as follows: Section 2 discusses related work, Section 3 describes The perspective-n-point (PnP) problem , Section 4 presents the experimental results, and Section 5 contains the conclusions.

2. RELATED WORK
The gradual advancements in technology will have a significant impact on different sectors of the economy. The development in 3D pose estimation is an example of the positive changes that modern technology will have in society. 3D pose estimation involves using a feature-based application to transform a 2D object to facilitate face recognition and character animation [7]. Research conducted in the last decade has not effectively looked at face detection and 3D face estimation [14]. Research into this field has been limited by the availability of technology and lack

of data for analysis. 3D pose estimation can be categorized into geometry and nonlinear regression methods. The work involved has both 2D and 3D methods. The algorithms are based on 2D images and include facial-feature-based approaches (scale-invariant feature transformation) and full-face appearance-based methods such as eigenfaces and direct correlation techniques [15]. 2D images rely on matrices and not 1D vectors, which is beneficial because there is no need to transform them before extracting features. This can be seen in the incorporation of a camera pair to calculate a three-layer neural network and depth but to estimate head pose angles to detect skin color by Seeman [16], [17]. The challenge of using an external camera pose is that the camera remains static when tracking the heads movement [18], [19]. The use of these algorithms improves efficiency, as they create a refined image by focusing on features and the whole figure. Manifold learning, locally linear embedding, and locality preserving projections are other algorithms that help to facilitate 3D face estimation [20]. These algorithms carry the assumption that a low-dimensional manifold gives variation in head poses [21]. Lanitis et al. adopted a robust search to match feature points by extracting facial features in an active shape model [22]. These algorithms work with various resolutions, slowly looking for the learned shape, ranging from a coarse resolution to the finest resolution possible [23]. Trivedi and Murphy-Chutorian tested various algorithms applied during HPE in two ways that include extraction and estimation of the head pose [24]. Their work helped refine the results because it improved efficiency and successfully tracked the head. They achieved this by using feature-based detectors to assist in the extraction of head features to estimate the pose while sensors dependent on appearance used the entire face without adding features [25]. These algorithms set the head pose of the face used during input as the 3D angles of the most familiar template [3]. Their work has been used by other scholars to shape the direction of their studies and methodologies. Cipolla and Gee used five significant elements to estimate facial orientation, such as nose tips, points, outer-eye corners, and mouth corners [3]. Fanelli et al. calculated the probable location of the head pose and nose tip [8]. Li and Pedrycz realized automatic nose detection using curvature information such as principle and mean curvature [26]. These scholars focused on improving the quality of features and making objects realistic in the 3D face estimation process. The heads motion patterns, as well as the orientation, are important non-verbal cues that produce valuable data concerning the behavior and attitude of a person, which help in 3D HPE [27]. Yang used an RGB image simultaneously corresponding to depth with the aim of locating the pose of the human head [28]. Wang et al. calculated the probable pose of the head from the RGB pictures gathered using a Kinect sensor [29] [30]. To instruct a classifier, a 3D morphable model to engineer the head pose is used in combination with a collection of about 50K images [31]. The Kinect SDK offers various functions for depth map capture as well as body skeleton tracking [32]. The different researchers who have contributed to the field have been improving on the foundation laid by other scientists. This can be seen in the fact that works by Trivedi and Murphy-Chutorian were followed by an increased focus on individual features such as point of mouth, eyes, and nose. The contributions by researchers like Seeman and development of algorithms by Keskin and Kondori et al. have been vital in the continuous growth and development of the field. The advancements in technology will help to facilitate better algorithms and techniques to be used in 3D face estimation in the future [33]. Efforts should be made to develop an algorithm that will detect changes in face patterns and use these triggers to communicate. This will help to improve interactions between man and machine in society.

3. THE PERSPECTIVE-N-POINT (PNP) PROBLEM

The perspective-n-point (PnP) problem is defined as the problem of estimating the pose of a calibrated camera, given three-dimensional points in world coordinates and their respective two-dimensional projections, onto the image plane. The term pose is used to describe the six degrees of freedom comprised of the three rotation angles (roll, pitch, and yaw) and the three translation parameters of the camera. More precisely, if the three-dimensional points in world coordinates are denoted as

$$\begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix}, k = 1, 2, ..., n$$

and the respective two-dimensional points / projections onto the image plane as

$$\begin{bmatrix} u_k \\ v_k \end{bmatrix}, k = 1, 2, ..., n$$

then the PnP problem is defined in terms of the equation

$$s \begin{bmatrix} u_k \\ v_k \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & y & u_o \\ 0 & f_y & v_o \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \\ z_k \\ 1 \end{bmatrix}$$

where the scope is the computation of the rotation parameters t_{ij},i,j = 1,2,3 and the translation parameters t_i,i = 1,2,3. In the previous equation, S is a scaling factor and the matrix

$$K = \begin{bmatrix} f_x & y & u_o \\ 0 & f_y & v_o \\ 0 & 0 & 1 \end{bmatrix}$$

is the matrix of intrinsic camera parameters, which is assumed to be estimated via a camera calibration procedure. $f_x$and $f_y$are the scaled focal lengths along each dimension, y is the skew parameter, and (u_o,v_o) defines the so-called principal point of the camera.

4. EXPERIMENT

In this section, experimental methods used to evaluate the performance of the proposed methods for detecting and tracking faces in 3D using stereo camera arrangements is provided. The experiment sections are organised as follows: section IV-A describe the dataset and experiment tools; section III describes the camera calibration; section 4 describe presents experimental results.

4.1 Dataset

To the best of the authors knowledge, no stereo dataset for geometries that are similar to hand-held smart devices is freely available online for academic use. Since two USB cameras were used to track the faces stereo images and videos were obtained. The face data were in an environment which the lighting, illumination, and other factors were controlled. The baseline between the two cameras varied from 5cm to 13cm, and object were located at different distance from the cameras.

1) Experiment tools : In order to conduct the experiments, a small laboratory was established. It contained the following pieces of equipment. 1. Raspberry Pi 2 Model B: The Raspberry Pi 2 is a single-board computer uses the Linux operating system. It was developed in the United Kingdom by the Raspberry Company. The Raspberry Pi2 has a quadcore processor thats runs at 900 MHz with 1 GB of RAM , 4 USB ports, video and audio outputs . 2. Two USB Raspberry cameras: The cameras were connected to the Raspberry Pi 2.

4.2 Camera Calibration and Intrinsic Camera Parameters

A more common model with reference to a 3D-scenecapturing method that rests upon using a camera also gives considers the distortion due to a non-ideal display lens. The current extended approach is examined based on the following equations:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + t$$

$$x^0 = x/z \; y^0 = y/z \; r2 = x02 + y02$$

$$x'\prime = x' \frac{1 + k_1 r^2 + k_2 r^4 + k_3 r^6}{1 + k_4 r^2 + k_5 r^4 + k_6 r^6} + 2_{p1} x' y' + p_2(r^2 + 2x'^2)$$

$$y'\prime = y' \frac{1 + k_1 r^2 + k_2 r^4 + k_3 r^6}{1 + k_4 r^2 + k_5 r^4 + k_6 r^6} + 2_{p1} x' y' + p_2(r^2 + 2x'^2)$$

$$u = f_x x'\prime + c_x$$

$$v = f_y y'\prime + c_y$$

In the aforementioned equations, $k_1$through k_6 posit as radial distortion coefficients, whereas p_1 and p_2 serve as tangential distortion coefficients. Each of these coefficients needs to be assessed throughout a calibration phase oriented towards combining the captured images by the camera (e.g., a checkerboard) and the assessed coefficients with the intent to correct camera mistakes. Consequently, the cameras matrix by means of utilising images that reflect various camera calibration patterns has been assessed. A traditional way of producing a calibration pattern consists in printing out a checkerboard image, such as those that can be found in Fig. 2. Utilising a set of this sort of images, the computation of intrinsic/extrinsic camera parameters for each of the two cameras was performed, which in turn proved to be identical for both cameras.

$$A_1 = \begin{bmatrix} 1400.45 & 0 & 600 \\ 0 & 1400.45 & 250 \\ 0 & 0 & 1 \end{bmatrix}$$

for the first camera and

$$A_2 = \begin{bmatrix} 1399.41 & 0 & 550 \\ 0 & 1399.41 & 274 \\ 0 & 0 & 1 \end{bmatrix}$$

Fig. 1 : Instance of the distortion caused by camera lens. IIn lieu of producing an image deprived of any distortion, a camera often produces an image that has much in common with a mirrored outer surface of a section of a sphere (middle image) or a mirrored inner surface of a section of a sphere (right image). The assessment of the parameters k_1, through k_6,along with p_1 and p_2, is likely to help determine the form of the non-linear lens distortion. Hence, by revealing the characteristics of those parameters, lens distortion can be fixed.

Similarly, for another camera, the lens distortion parameters were determined based on the same characteristics: :

$$\begin{bmatrix} k_1 & k_2 & p_1 & p_2 & k_3 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.21703 & 0 & 0 & 0 \end{bmatrix}$$

for the first camera and:

$$\begin{bmatrix} k_1 & k_2 & p_1 & p_2 & k_3 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.22908 & 0 & 0 & 0 \end{bmatrix}$$

for the second camera. The present parameters were then converted into rotation and translation coefficients. Extrinsic parameters such as rotation and translation vectors also indicate the transformation that plays a crucial role in finding locations of the objects in the world coordination space. The basic values for the camera's rotation vector and camera's translation vector are as follows :

$$\begin{bmatrix} r \end{bmatrix} = \begin{bmatrix} 0.007848 \\ 0.006589 \\ 0.000576 \end{bmatrix}$$

and:

$$\begin{bmatrix} T \end{bmatrix} = \begin{bmatrix} -120 \\ 0 \\ 0 \end{bmatrix}$$

It cannot be overstressed that the camera's rotation vector may be converted into a rotation matrix, and emphasis is placed on equation 1 to perform this operation.

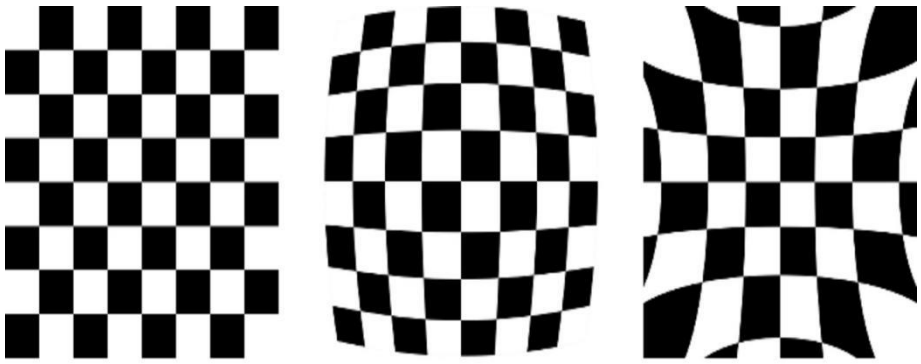
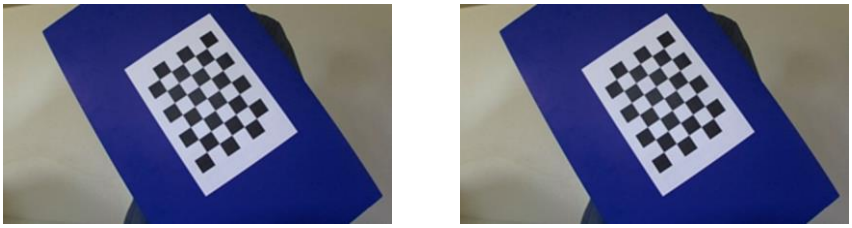


Figure 1. Instance of the distortion caused by camera lens.



(a) left Camera (b) Right Camera

Figure 2. Example checkerboard used to compute the parameters of the left and right cameras.

4.3 Stereo Rectification

Then, the camera matrices for each of the cameras, the lens distortion parameters for *A*1 and *A*2, and the translation vector T and camera's rotation matrix*R* were computed, as previously mentioned. Next, an optional rectified pair *R*1 and *R*2 as well as two projection matrices, such as *P*1 and *P*2, in the new project coordinate system were computed. In this survey, a horizontal stereo setup where both camera angles are shifted relative to each other along the x-axis found its application. Based on this case, the projection matrices incorporate the following forms:

$$P_1 = \begin{bmatrix} f & 0 & cx_1 & 0 \\ 0 & f & cy & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

and:

$$P_2 = \begin{bmatrix} f & 0 & cx_2 & T_x * f \\ 0 & f & cy & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

whereT_x must be seen as a horizontal shift of images between the two cameras. The initial three columns of the projection matrices P1 and *P*2 are expected to become new camera calibration matrices. They, along with the pair *R*1 and *R*2, were submitted to the initUndistortRectifyMap [34] function to establish the rectification maps, and the given operation was performed separately for both cameras.

For the scopes of pose-estimation, we are interested in the new, rectified camera matrix for the left camera. This matrix is given by the first three columns of matrix P_1, and it is in effect the required *K* matrix. In the setting examined, this matrix was found to be equal to :

$$K = \begin{bmatrix} 1291.04 & 0 & 592.73 \\ 0 & 1291.04 & 261.23 \\ 0 & 0 & 1 \end{bmatrix}$$

As it will be clearer in the sequel, the rectified camera matrix of the left camera is required. The reason for this is that, as it will be described in Subsection 4.5, depth estimation is performed for points that appear on the frame of the left camera

4.4 Computation of the Image Points Via Tracked Points

A discriminative facial deformable model was trained by a cascade of regressors through videos. The model was developed in [3] as a computationally efficient method for updating the deformable model, and it can be used to track several landmark points of the human face from frame to frame. The tracked algorithm was applied to both the left and right camera images after rectification and stereo calibration, because subsequent depth estimation was performed for the points of the undistorted images. Thus, since the 2D points must correspond to the 3D points computed in the next subsection, the 3D face tracker was applied to the undistorted images. The facial landmarks points detected and tracked are denoted as

$$P_L^{(t)}(K) = \begin{bmatrix} u_k^{(t)} \\ v_k^{(t)} \end{bmatrix}, k = 1, 2, ..., n = 49$$

for the left camera frame and

$$P_R^{(t)}(K) = \begin{bmatrix} q_k^{(t)} \\ p_k^{(t)} \end{bmatrix}, k = 1, 2, ..., n = 49$$

for the right camera frame, at discrete time *t*, which Figure 3 shows the points model.

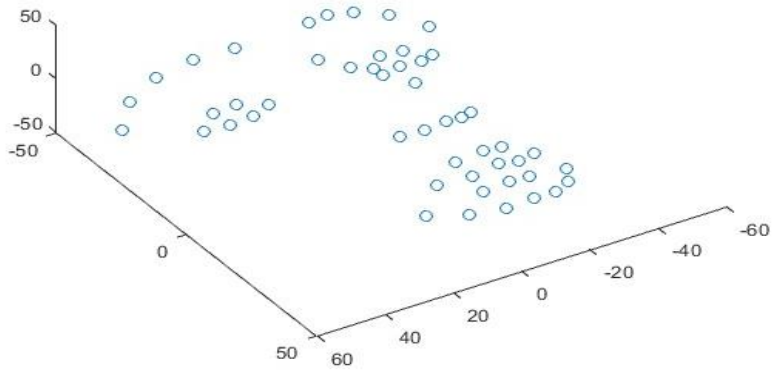


Figure 3. Model points position.

4.5 Computation of the 3D World Points

As already mentioned in the previous section, pose estimation requires the coordinates of the 3D points in the world frame of reference. This, in turn, requires the extraction of depth information, and computing depth information using a stereo camera arrangement is traditionally performed via the so called disparity map computation. More precisely, after calibrating the left and right cameras and correcting any nonlinear distortions, the disparity map of the left frame can be computed, and the disparity map is an intensity (e.g. grayscale) image, where for each pixel on the left image, the same pixel is localised on the right image. Their distance is computed, and this is the disparity for this pixel. The disparity value is linearly mapped to an intensity to generate the disparity map, and the disparity of a pixel is inversely proportional to its corresponding depth. The previous computation of the disparity map may not be accurate due to several reasons. For example, not all pixels appear on both the left and right images. A more accurate estimation of the disparity of certain points can be computed if these points have been localised on both the left and right frames. In fact, this task can be performed since the 3D face tracker has been applied to both the left and right camera frames. Thus, the disparity that corresponds to these key landmark points is given as follows:

$$D^t(k) = \left\| P_L^{(t)}(K) - P_R^{(t)(k)} \right\|_2, k = 1, 2, ..., n = 49$$

where $P_L^{(t)}(K)$ and $P_R^{(t)}(K)$ were defined in the previous subsection, and they correspond to the key landmark points of the left and right frames computed by the face tracker at time (frame number) *t*. Having computed the disparity for point *k* at time *t*, as shown above, the *Q* matrix computation can be used to compute the 3D coordinates for these points. In particular,

$$Q = \begin{bmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 0 & f \\ 0 & 0 & -\frac{1}{T_x} & \frac{C_x - C'_x}{T_x} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -592.73 \\ 0 & 1 & 0 & -261.23 \\ 0 & 0 & 0 & 1291 \\ 0 & 0 & 0.0083 & 0 \end{bmatrix}$$

and depth estimation is performed using the equations :

$$\begin{bmatrix} X_k'^{(t)} \\ Y_k'^{(t)} \\ Z_k'^{(t)} \\ W_k'^{(t)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -c_x \\ 0 & 1 & 0 & -c_y \\ 0 & 0 & 0 & f \\ 0 & 0 & -\frac{1}{T_x} & \frac{C_x - C'_x}{T_x} \end{bmatrix} \begin{bmatrix} u_k^{(t)} \\ v_k^{(t)} \\ D^{(t)}(k) \\ 1 \end{bmatrix}$$

and finally, the 3D point that corresponds to $P_L^{(t)}(K)$ is given as

$$X_k^{(t)} = \frac{X_k'^{(t)}}{W_k^{(t)}}$$

$$Y_k^{(t)} = \frac{Y_k'^{(t)}}{W_k^{(t)}}$$

$$Z_k^{(t)} = \frac{Z_k'^{(t)}}{W_k^{(t)}}$$

4.6 Experimental Results

Our dataset consists of 30 videos with 300 frames per video that can be used to test the proposed 3D-face tracking. Figure 4 shows the determination of face orientation in the form of yaw, pitch, and roll angles. Pitch and roll angles are very similar on both sides, but yaw angles have offsets relative to the other camera. That is expected, because the orientation of the face around the vertical axis is the most differently seen from both cameras, while the other two angles should be similar in views from both sides. In agreement with the fact that the pose of an object varies slowly with time, Figure 5 shows the expected smoothest curves of 3D algorithms obtained from stereo video, while Figure 6 shows the face positions in a matter of depths. For each image of our dataset, head pose parameters were calculated using the proposed method. The head pose parameters contain orientation parameters (yaw, pitch, and roll) and positions parameters, and our dataset was labelled by taking a physical front face model to calculate the average angle. Lastly, a low rate of average angle error ~1.9 was achieved compared to the state-of-art methods [35], resulting in an accuracy of 98.6

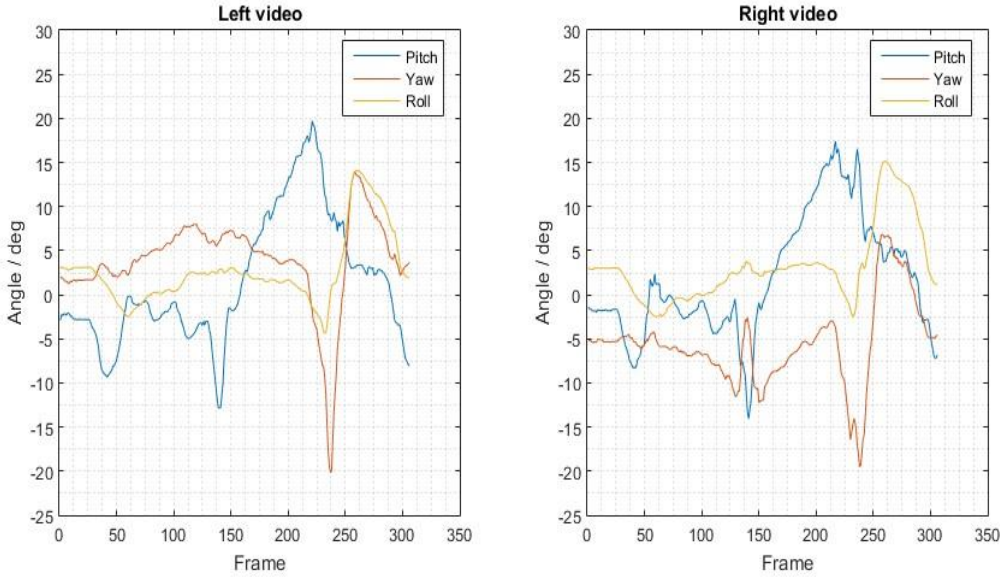


Figure 4. Orientation data for left and right frames.

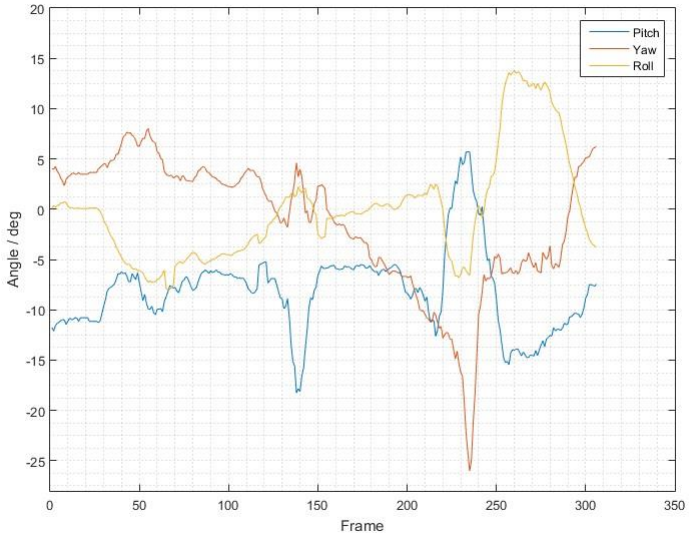


Figure 5. Orientation data from stereo video.

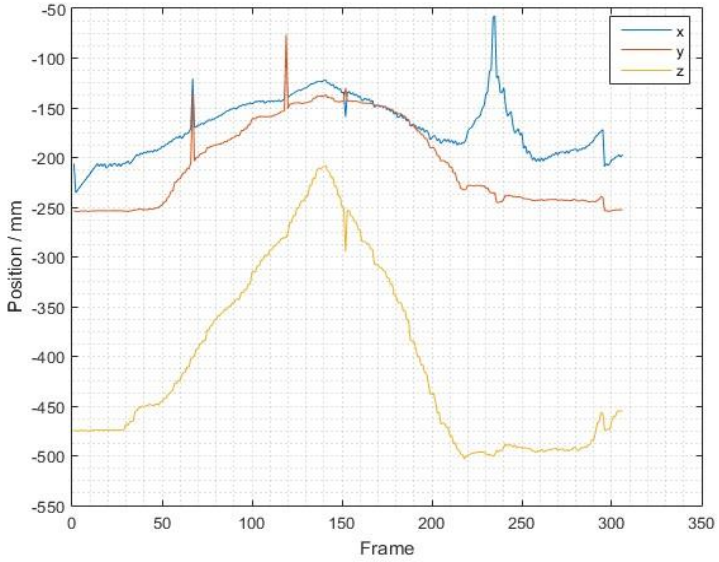


Figure 6. Face position.

5. CONCLUSION

In conclusion, problems related to the field of HPE should be solved in a structured manner. This field faces constant multidimensional improvements. It was already proven that the limitations of pose estimation require strategies based on 3D techniques. Currently, it is possible to achieve high-quality images using several cameras. The images are not blurred, but a new challenge arises. As various HPE methods exist, all of them require further development and investigation. The PnP problem is one of the most important to solve, as it allows researchers to establish an accurate correlation between 3D coordinates in the world and 2D projections. The solution of the PnP problem would help researchers estimate the position of the camera responsible for obtaining the correlation. In this paper, a method for head pose estimation using a stereo camera arrangement was implemented. The method follows a route alternative to using a 3D model of the head and a single camera, namely, using a calibrated stereo camera without a 3D model. The method developed in this work can detect and track several landmark points of the human head using videos. By applying this algorithm at the left and right camera frames, accurate disparity can be computed for these landmark points. Thus, the 3D coordinates for these points can be computed. Finally, the 2D and 3D points can be used to create an instance of the PnP problem and to perform pose estimation.

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